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Transgenic tilapia and the tilapia genome

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Received 10 August 2001; accepted 18 May 2002

Abstract

The tilapia fish (*Oreochromis niloticus*) has an important place in the aquaculture of the developing world. It is also a very useful laboratory animal, and readily lends itself to the transgenic technology. Through the use of reporter genes, a range of potential gene promoters have been tested in tilapia, both through transient and stable expression of the reporter construct. Using the transgenic technology, growth enhanced lines of tilapia have been produced. These fish have no abnormalities and offer a considerable growth advantage for future exploitation. It is however crucial that transgenic fish, to be exploited in aquaculture, be sterile, and various methods of achieving sterility are considered. These include triploidy, gene knock out of crucial hormone encoding genes via homologous recombination, and knock down of the function of the same genes via ribozyme or antisense technologies.

Transgenic tilapia also offer the potential for exploitation as biofactories in the production of valuable pharmaceutical products, and this is also discussed. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Transgenic fish; Reporter gene; Antisense; Ribozyme; Biofactory

1. Introduction

Tilapia fish are of huge importance in world fisheries, being the second most important group of food fishes in the world, next to the carps. There are about 100 species and subspecies, of which *Oreochromis niloticus*, *Oreochromis mossambicus* and *Oreochromis aureus*, and hybrids between them, are much the most important. *O. niloticus* accounted for a harvest of nearly 0.9 million tonnes in 1998 (FAO, 1999), second only to carp as a warm water food fish and exceeding the harvest of Atlantic salmon, *Salmo salar*, although the value of the Atlantic salmon catch is more than twice that of the tilapia catch.

Although native to Africa, tilapias are cultured in much of Asia and the far East, and really occupy two rather separate market niches, being a poor man's food fish in much of Asia and Africa, but being a high value food fish in countries such as Israel and the Southern United States. Tilapias are omnivorous, but will subsist well on a vegetarian diet, are rela-

tively robust and disease free, and are capable of rapid growth, being sexually mature at 6/7 months from hatch and marketable at this age. The male fish grows faster and becomes larger than the female and is the more desirable fish at market.

O. niloticus is also an excellent laboratory animal and deserves to be better known as such. It is a mouth-brooder, and in aquarium conditions is reproductive throughout the year. Eggs and milt can be stripped from ripe fish, fertilised by hand, and the eggs reared to hatch in small Zugler vessels. Perhaps the greatest drawbacks of the fish as laboratory model animals is their aggressivity – individual brood stock are best kept in isolation – and in genetic terms, knowledge of the tilapia genome is rather preliminary, and far behind *Fugu* and the zebra fish. Table 1 lists the particular benefits and limitations of tilapia as a model species. It must also be emphasised that fish are ancient and very diverse in evolutionary terms, so the molecular sequence homology between different teleost families is often less than between other vertebrate classes.

2. Materials and methods

Since this paper is in many respects a review of work, much of which is already in publication, this section will be brief.

Abbreviations: LUC, firefly luciferase; GFP, green fluorescent protein; NEO, neomycin resistance; G0, first generation transgenic fish; G1, second generation transgenic fish; MUG, methyl umbelliferone galactoside; GH, growth hormone; GnRH, gonadotropin releasing hormone; LH, luteinising hormone; PCR, polymerase chain reaction

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Table 1

Tilapia (*O. niloticus*) characteristics as laboratory model species

1. Warm water required, since fish will not survive much below 11°C, and 28–30°C is optimal (Ross, 2000)
2. Fresh water required, although *O. niloticus* will survive in mildly brackish water. *O. mossambicus* is more tolerant of high salinity than is *O. niloticus*
3. Reproductive throughout the year, males being constantly capable of being stripped, and females spawning approximately every 3–4 weeks. Eggs can be stripped from females only minutes before natural ovulation, so it is necessary to recognise when a particular female is ripe by appearance of genital papilla and 'nest' building behaviour with gravel
4. Eggs, available as a few hundred per ovulation, can be reared after fertilisation in small Zugler vessels in which the eggs are constantly agitated by a water current. Hatching occurs in 72–96 h, and embryos are nutritionally dependent on the yolk sac until approx. day 10
5. Very robust and disease free
6. Adults remain reproductive for about 2/3 years
7. Individuals are very territorial and aggressive and so brood stock fish must be segregated
8. Adult fish are large, up to 25 cm in length at 8 months, and so a large aquarium system is needed

2.1. Transgenic fish used

The transgenic fish described were produced by micro-injection of fertilised eggs as discussed in Rahman and Maclean (1992). Following injection of linear transgene molecules into the egg perinuclear cytoplasm, eggs and embryos were kept in mini-Zugler vessels until yolk sac absorption, with a constant upward current of water to keep the eggs and embryos agitated. Positive transiently expressing fish were detected by assays for the reporter gene expression, while fish with stable integrated transgene copies were detected by initial polymerase chain reaction (PCR) of fin clips and subsequent Southern blots.

2.2. Growth enhanced tilapia

The growth enhanced tilapia tested in Szarvas, Hungary carried a single copy of a chinook salmon (*Oncorhynchus tshawytscha* Walbaum) growth hormone gene spliced to an ocean pout (*Macrozoarces americanus* Black and Schneider) antifreeze promoter co-ligated with a carp beta actin/ beta galactosidase (lacZ) reporter gene construct, integrated into the tilapia genome. The salmon growth hormone gene construct was kindly given to us by Prof. Choy Hew. Details of the growth trials of these fish will be found in Rahman et al. (2001).

2.3. Detection of lacZ by methyl umbelliferone galactoside (MUG) assay

The MUG assay was carried out by the method described on Hoefer's Technical Bulletin #129 by using a Hoefer TKO 100 minifluorometer. In the MUG assay, 4-methyl umbelliferyl- β -D-galactoside was used as a substrate which was then cleaved by β -galactosidase to yield the fluorescent molecule 4-methylumbelliferone (7-hydroxy-4-methylcoumarin, MU). After excitation at 365 nm, 4-methylumbelliferone emits light at a wavelength of 460 nm. This assay is sufficiently sensitive to detect picogram quantities of β -galactosidase.

3. Results and discussion

3.1. Transgenesis in tilapia

Transgenic fish are useful for a number of different reasons. They can provide good models for studies on gene regulation and gene expression during development, transgenesis provides a way of improving fish for aquaculture by conferring desirable genetic traits, and transgenic fish can be exploited as biofactories to produce valuable pharmaceuticals.

Fish of many species have been used in transgenesis, for example common carp (*Cyprinus carpio*) by Zhang et al.

Table 2

Benefits and limitations of tilapia fish as a model organism for the transgenic technology

Benefits

1. Short generation time of 6–7 months
2. Eggs constantly available in a laboratory system
3. Embryos and fry easily reared
4. Embryos semi-transparent, so some reporter genes can be detected without sectioning or sacrifice
5. Eggs fairly easily injected via micropyle, i.e. easier than eggs of salmonids, although the egg is opaque and the chorion tough. However, injection needles must be mounted in a micromanipulator and each egg individually positioned for optimal needle entry
6. The increased size over medaka and zebrafish means that blood and tissue volumes and sizes are much increased

Limitations

1. Embryos less transparent than those of zebrafish, and so transient expression less easily detected with reporter genes
2. Knowledge of genome and early development is very limited as compared to zebrafish
3. Eggs less easily injected than those of zebrafish, medaka and carp

(1990), channel catfish (*Ictalurus punctatus*) by Dunham et al. (1987), African catfish (*Clarias gariepinus*) by Volckaert et al. (1994), Northern pike (*Esox lucius*) by Guise et al. (1992), rainbow trout (*Oncorhynchus mykiss*) by Guyomard et al. (1989), Atlantic salmon (*S. salar*) by Shears et al. (1991), mud loach (*Misgurnus mizolepin*) by Nam et al. (2000), and smaller model species such as goldfish (*Carassius auratus*) by Yoon et al. (1990), and zebra fish (*Danio rerio*) by Stuart et al. (1990).

Transgenic tilapia have been produced by Brem et al. (1988) and Martinez et al. (1996) and by Rahman and Maclean (1992), but the Havana, Cuba laboratory and the Southampton, England laboratory are now the chief centres working on transgenesis in tilapia. Tilapia have much to offer as a transgenic system and the pros and cons are listed in Table 2. Particularly noteworthy are the short generation time, constant egg availability, and comparative ease of egg injection and rearing of the embryos. Thus the generation time is about 6 months, little longer than zebra fish, and the rearing of the fry is substantially easier than with zebra fish. However, the embryos are somewhat less transparent than

those of zebra fish, and the depth of knowledge about the genome and aspects of molecular development are much less than in that species. Although egg injection is not as easy as with carp or zebrafish, it is much easier than with salmonids.

Fish have their own characteristics as a transgenic system, and these hold true for tilapia transgenesis. Table 3 outlines the major features.

3.2. Use of reporter genes in transgenic tilapia

Reporters such as chloramphenicol acetyl transferase (CAT), lacZ, firefly luciferase (LUC), and green fluorescent protein (GFP) have been widely used in fish (see Section 3 in Iyengar et al., 1996; Maclean, 1998). The neomycin resistance reporter/selection system is less useful since the mosaic distribution and expression in the first generation transgenic fish (G0) embryos and fish means that the embryos or fry are killed by the neomycin or G418 since non-transgene-expressing cells are susceptible. Good quantitative assays for GFP expression are difficult to set up,

Table 3
Transgenesis in tilapia fish (many aspects cited hold true for all transgenic fish systems)

1. Although many alternatives to needle microinjection of the fertilised egg have been tried, such as electroporation, gene guns, oocyte injection and liposome mediated gene transfer via sperm, *none have, to date, proved reliably superior to microinjection*
2. The fertilisation pronuclei cannot be visualised, even microscopically, so *DNA copies have to be injected into the perinuclear cytoplasm* of the fertilised egg. Subsequent uptake into cell nuclei and chromosomal integration is therefore not guaranteed
3. *Many transgene copies are required, 5×10^5 in tilapia*. This is a balance between injecting too much DNA, which is lethal to egg and embryo survival, and injecting a low copy number and finding that gene integration is exceedingly infrequent
4. Following embryonic development, *a majority of embryos will display transient expression* of some transgene copies. This is from transcription and translation of unintegrated transgene copies. Although the transient expression is mosaic (see 5 below) it is tissue specific and dependent on the characteristics of the promoter driving the construct. However, transient expression levels are highly variable from embryo to embryo, and so large numbers of embryos must be screened if data is to be quantitative
5. *Expression in the first generation fish* grown from injected eggs *is invariably mosaic*, presumably because the transgene copies are distributed in a somewhat random way following early cell division in the embryo. Stable expression (see 6 below) in first generation fish is, like transient expression, invariably mosaic
6. Only a small proportion (1–5%) of injected eggs will produce fish showing stable transgene expression from integrated copies
7. Since even fish with integrated transgene copies have a mosaic pattern of integration, *only a subset of G0 fish will demonstrate germ line transmission*
8. Once second generation transgenic fish (G1) progeny are produced, the problem with mosaicism is solved, but the *proportion of G1 positive transgenics may be low*
9. PCR from fin clip material can be used to identify presumptive transgenic positives, but some will be missed because of mosaicism. *PCR should be followed up with Southern blotting to confirm the presence of the transgene*. Some long term persistence of non-integrated copies is possible, so Southern blotting and germ line transmission are needed to confirm chromosomal integration
10. *All transgenic fish are initially hemizygous*, that is that integration occurs on only one of a pair of homologous chromosomes. However homozygous fish can be produced by crossing two hemizygous fish of the same line and identifying the 25% homozygous progeny
11. *Transgenes readily form concatamers in fish*, and so most transgenic fish will be found to have multiple copy integration events. Multiple integrations into separate chromosomes are also not uncommon
12. As will be discussed in a later section, *there is no equivalent to the mouse embryonic stem (ES) cell system in fish*, so gene knock-out and homologous recombination is presently difficult
13. *A range of reporter genes* such as CAT lacZ LUC and GFP are useful in tilapia, and some, such as GFP, can be detected in living fish, although expression is hard to quantify precisely
14. A trained worker producing transgenic tilapia should be able to reach the following targets
 - (a) Eggs injected per day –50
 - (b) Eggs hatching to yield viable embryos after egg microinjection –60 out of 100
 - (c) Eggs developing to show transient expression between 2 and 10 days of embryonic development –40 out of 100
 - (d) Eggs developing into embryos and yielding fish showing mosaic stable expression at 1 month, –1–5 out of 100
 - (e) Fish grown from injected eggs which demonstrate some germ line transmission –0.1–1 out of 100 eggs injected
 - (f) Percentage of positive transgenic progeny from a cross between a stable G0 transgenic and wild type fish –0.5–10 out of every 100
 - (g) Percentage of positive transgenic progeny from a cross between a stable G1 transgenic and a wild type fish –50 out of 100 (i.e. hemizygous)

although it has the merit of being detectable in living embryos. LacZ has much to commend it and the MUG assay provides a sensitive way to measure expression levels, but unfortunately there is evidence that in some cells and tissues, in fish, as in mice, (Montoliu et al., 2000) the reporter is capable of silencing the promoter in use and so some degree of transgene expression mosaicism results. (Rahman et al., 2000).

The following experiments have been conducted in tilapia using reporter gene transgene constructs.

3.2.1. Demonstration of copy number related expression levels in different transgenic lines

As demonstrated in Rahman et al. (1997) three different transgenic lines were compared, in which the concatamered copy numbers were 19, 34 and 330 copies in each of the C86, C118 and C58 lines respectively. Mature fish from each line were dissected and the brains removed for subsequent lacZ expression level assay. As can be seen in Fig. 1, the expression levels are broadly in line with the copy number. The copy number was determined by densitometric determination following Southern blotting on gels. A more detailed quantitative assay was carried out by MUG assay of homogenised G2 and G3 non-mosaic embryos of each line. In each case, when the expression level in the homogenates (expressed as $10^3 \times \text{MU units}/\mu\text{g protein}$) were divided by the determined copy number, figures of 0.29, 0.32 and 0.23 were obtained for the respective lines C86, C118 and C58, suggesting good correspondence. This evidence is cited not to suggest that such a correlation will always obtain in fish – almost certainly it will not – but to emphasise the usefulness of the reporter gene assay.

Use of reporter gene constructs to optimise promoter selection for subsequent inclusion in new gene constructs or to demonstrate expected patterns of tissue specific expression.

When new promoters are chosen to drive particular transgene sequences, it is crucial to determine what length of upstream regulatory sequence will give maximal expression. This is usually a matter of including any available enhancers and avoiding any silencer elements. Such assays may of course be readily carried out in tissue culture cells where the promoters are ubiquitously expressed, but careful analysis of transient or stable expression in fish is often necessary when no relevant tissue cell line is available. Thus, we have recently helped to characterise a tilapia L18 ribosomal protein gene by testing promoter regions of differing lengths in CV1 cells (monkey kidney cell line), and by measuring transient expression in both tilapia and zebrafish embryos (Molina et al., 2001), in combination with a lacZ reporter gene. A tilapia heat shock HSP70 gene has also been isolated and characterised, (Molina et al., 2000), although in this case the expression analysis was carried out with transient expression in zebrafish embryos, rather than with tilapia embryos.

3.2.2. Use of reporter gene constructs to demonstrate transgene homozygous integration status

It is sometimes desirable to obtain fish in which both copies of a homologous pair of chromosomes carry integrated transgenes. In a recent paper (Rahman et al., 2000), we have demonstrated the use of the lacZ reporter to indicate the production of homozygous transgenics by crossing

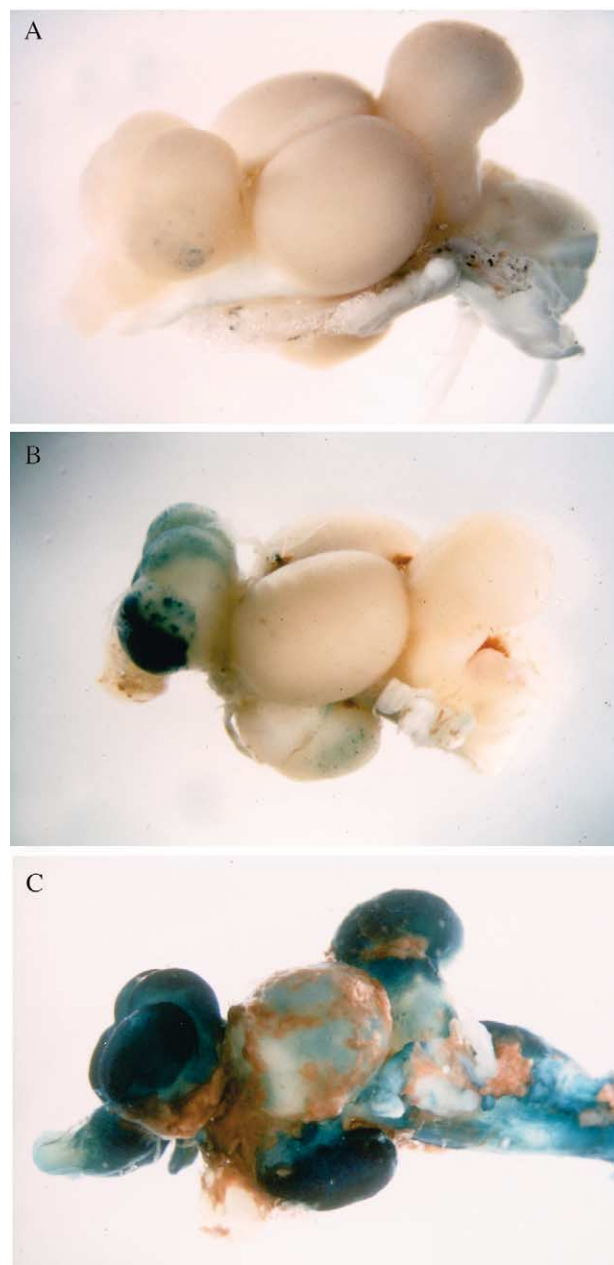


Fig. 1. (A–C) Brains of 7 month old G2 fish of the C86, C118 and C58 lines, respectively, following dissection and staining after removal with X-gal (magnification $\times 10$). Note the strong expression in one area of the telencephalon (left of the picture) in all three lines, contrasting with the limited expression in the cerebellum and optic tectum evident only in the C58 line (C). Copy numbers of the reporter gene concatamer in the three lines are 19, 34 and 330, respectively. (Reproduced with permission from Transgenic Research).

together two hemizygous transgenics of the same line. The authentic transgene homozygosity was subsequently proven by showing that, when crossed with a wild type tilapia, 100% of hemizygous transgenic progeny were obtained.

3.3. Use of transgenic technology to produce improved lines of tilapia for exploitation in aquaculture

It has been clear for some time that the transgenic technology could be used to produce fish with improved genetic traits such as faster growth, improved cold tolerance or freeze resistance, increased disease resistance, or enhanced tolerance of salinity. In this paper the production of growth enhanced lines of tilapia will be used to illustrate this aspect.

3.3.1. Production of growth enhanced lines of tilapia

Ever since Palmiter et al. (1982) published their results showing dramatic growth enhancement in transgenic mice, fisheries biologists have known that a parallel success was possible in fish. However, the most impressive results in transgenic fish were not published until 1994, when Devlin et al. (1994) published the results of their experiments with Pacific salmon. In salmonids, using heterologous growth hormone genes spliced to strong promoters of fish origin, it proved possible to show more than 10-fold growth enhancement even with G0 fish, as compared to non-transgenic siblings.

Since production of growth-enhanced lines of fish is clearly directed at the goal of improving species for use in aquaculture, it is obvious that gene constructs should be designed with this in mind, avoiding as much as possible the use of sequences of viral origin and using whenever possible sequences from the same or related fish species. The design of growth enhancing gene constructs has followed to a great extent the experimental plan originally pioneered by Palmiter et al. (1982), namely to use a growth hormone coding sequence from a species closely related to the target animal, and to drive this with a promoter which will ensure synthesis in a large organ such as liver, or with a ubiquitous promoter ensuring widespread growth hormone (GH) synthesis.

In tilapia the work has been concentrated in two laboratories, that of de la Fuente in Havana, Cuba (Martinez et al., 1996) and our own laboratory in Southampton, England. The Cuban work has used a tilapia GH complementary DNA (cDNA) driven by a human cytomegalovirus (CMV) promoter, whilst the Southampton work has used a gene construct originally made by Prof. Choy Hew in Toronto (now in Singapore) and kindly gifted to us. This consists of a GH cDNA from chinook salmon, driven by a promoter from an ocean pout antifreeze gene. (See Du et al., 1992 for details of the original construct and Rahman et al., 1998 for details of the use of the construct in tilapia).

The growth enhancement obtained in tilapia by Martinez et al. (1996) is approximately 2-fold (1.8 times control) when fish are 7 months old, but data from extensive growth trials

have not been published for these fish. There is evidence from recent reports that some of these lines of growth enhanced tilapia have been commercially exploited and made available for food in local markets (de la Fuente et al., 1999).

The results of the growth enhancement work done by the Southampton group have already been published (Rahman et al., 1998, 2001; Rahman and Maclean, 1999), at least as far as non-commercial trials are concerned, and the data will simply be summarised here. It is important to stress that a number of different lines of fish were produced, and all resulted from coinjection of the GH construct along with another sequence, carp beta actin promoter driving a lacZ reporter (Rahman et al., 1997). This coinjection strategy was devised in order to increase the integration frequency of the GH transgene, and, insofar as this experiment was concerned, it succeeded. Although three transgenic lines were produced with integrated copies of the chosen GH construct, only one, the C86 line, has been extensively tested. This line contains a single integrated copy of the GH construct (Rahman and Maclean, 1999) and was chosen partly because the level of heterologous GH produced was low (1.02 ng/ml of serum), and it was assumed that better all round performance would be attained with low level increase of GH than with dramatic increase. Certainly in large mammals dramatic over-expression of GH can lead to serious abnormality (Cameron et al., 1994).

Following laboratory-based trials with the C86 line (Rahman and Maclean 1999), more extensive trials were carried out at a Fisheries Station at Szarvas, Hungary, in the winter of 1997/1998. The choice of this site resulted partly from the availability of local expertise in running fish growth trials and partly from effective physical containment. The latter resulted from the availability of warm geothermal water (which was used via a heat exchanger to warm good quality water), surrounded by cold water in local lakes and rivers. This combination ensured that, in the unlikely event of fish escaping, they would not survive in the locally cold water which surrounded the site. The results of the trials are reported in Rahman et al. (2001) together with separate nutritional trials carried out on the same line of fish at the Institute of Aquaculture, Stirling, Scotland.

In summary, the Szarvas trial lasted for 7 months and involved approximately 400 transgenic and 400 non-transgenic fish, reared both separately and together. The overall difference in mean mass was 2.5 times (see Fig. 2), the transgenic tilapia having a mean mass of 653 g as compared to 260 g for the non-transgenic fish. No serious abnormalities were evident, and 80 transgenic and 80 non-transgenic fish were dissected at the end of the period to determine whether internal abnormalities might have arisen. None such were evident.

Other parameters of interest are that food conversion efficiency was 20% better in the transgenics, and a digestibility trial suggested that the transgenic tilapia were more efficient utilisers of protein, dry matter and energy. They also had a higher moisture content and lower ash and crude protein



Fig. 2. Photograph showing three 30 week old transgenic fish (left) and three 30 week old non-transgenic full sibling fish (right) from the C86 line. The average mass of the transgenic and non-transgenic fish in this group was 653 and 260 g, respectively. The fish illustrated were representative of the entire batch. (Reproduced with permission from Fish and Fisheries).

content and crude lipid content in the flesh than non-transgenics on the basis of a small number of fish analysed. Rather similar data to the above have been obtained in studies on growth enhanced transgenic salmon.

From the above evidence it may be concluded that transgenic growth enhancement does produce tilapia with desirable faster growing properties. The growth rates measured are almost certainly not maximal, since only one transgenic strain was studied in depth, and there is evidence to suggest that the transgenic fish were slightly underfed towards the end of the trial period. Of course tilapia growth rate can also be improved by conventional selection, but the best

recorded improvements obtained by non-transgenic genetic selection are of the order of 1.5–2.0 times (ICLARM Report <http://www.egiar.org/iclarm/gclams/reshigh.htm>) the yield as compared to local strains.

Our laboratory is now engaged in trying to produce new strains of transgenically growth enhanced tilapia without using reporter gene coinjection and use of ‘all-tilapia constructs’ with tilapia promoters driving a tilapia GH gene (see Fig. 3). Such new strains would therefore become recognised as autotransgenics rather than allotransgenics in the terminology suggested by Beardmore (1997).

It is obvious that no GM (genetically modified) fish can

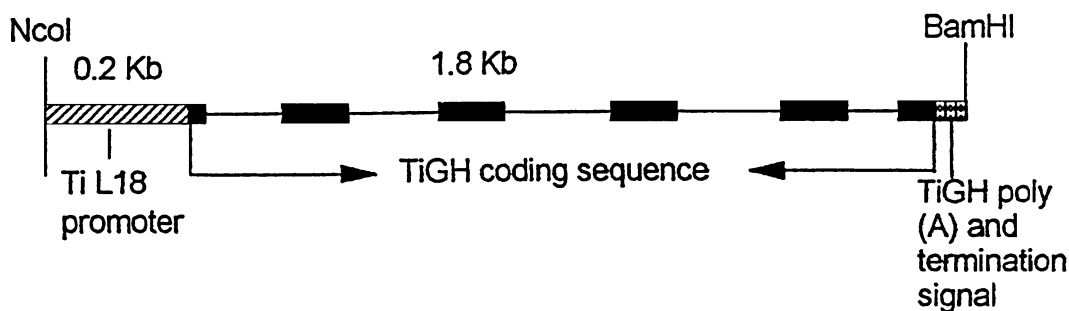


Fig. 3. Diagram of an ‘all-tilapia’ GH construct in which a tilapia GH coding sequence is driven by a tilapia L18 (ribosomal protein) promoter. Black boxes represent the exons. Thin lines represent the introns. Diagram is not to scale.

be safely exploited in aquaculture unless complete physical containment were available in an isolated land-locked lake, or if the fish could be biologically contained as a result of watertight sterility. This will be discussed in the following sections.

3.4. The need for sterility

As emphasised in the preceding section, sterility is a necessary adjunct to the exploitation of transgenic tilapia unless completely secure land locked facilities are available. In addition, sterility is an important parameter in its own right, even aside from its use for containment of GM fish. Tilapia itself is often a pest fish when released into large mixed fresh water fishery waters, since its tendency to precocious sexual maturation means that often large populations of small sized fish result, and indeed it may be a desirable warm water aquaculture species in situations where its establishment in the wild is undesirable. The problems posed by the escape of Atlantic salmon from sea cages and their interbreeding with wild populations in adjacent rivers are well known examples of problems with a different farmed fish species which could at least be partially reduced by the use of sterile fish.

Sterility in fish can be achieved by two different routes, namely ploidy manipulation to produce sterile triploids, or the use of transgenesis to achieve gene ‘knock-out’ or gene ‘knock-down’. Our laboratory is involved with all three approaches and these will be considered in turn.

3.4.1. Triploidy and tetraploidy in tilapia

Triploidy provides a way to render fish sterile but unfortunately triploid sterility is rarely 100% (Hussain et al., 1991). The lack of complete sterility is usually a result of some normal diploids hatching from treated eggs and some male triploids having partially functional testes and producing some viable sperm. This latter problem can be counteracted by using fish which are triploid and all-female through sex-reversal technology, but unfortunately, in tilapia, the male fish are larger and more marketable than the females and thus all-female triploids are an unattractive commercial proposition. Triploidy is normally induced in fish through temperature or pressure shock to the eggs preventing the extrusion of the second polar body (Purdom, 1993). As mentioned, another problem with such procedures is that often not all resulting fish are triploid and so even females can be fertile. An alternative strategy is to produce tetraploid fish through shocks at the first mitotic division and inducing a state of tetraploidy. Tetraploid fish, unlike triploids, are fertile, and produce all-triploid progeny on crossing with a wild type. Although tetraploid tilapia can be produced (see later), as discussed above the ‘leakiness’ of the triploid male makes the technology a less than complete answer to the question of sterility in tilapia.

Triploidy in tilapia is thus best seen as a partial but not complete answer to sterility. It thus reduces rather than

banishes fertility. If triploidy were to be used in this way to reduce the fertility of transgenically growth enhanced tilapia, then a further problem is encountered, since triploid transgenics may be actually less growth-enhanced than diploid transgenics (Razak et al., 1999). In these growth trials it was found that at 8 months of age normal non-transgenic tilapia averaged 39.9 ± 5.6 g, transgenic diploids of the C86 line averaged 174.9 ± 17.6 g while transgenic triploids of this line averaged 138.1 ± 9.7 g. However, it should be stressed that in this study all the triploids were from fertilised eggs resulting from sperm from a transgenic male being used to fertilise the eggs of a wild type female. The triploids therefore carried only one out of three chromosomes with an extra growth hormone gene. If the cross had been made the other way, which we propose to try in future, the loss of growth enhancement might be much reduced or absent altogether.

Hamid Farahmand in our group has undertaken the production of tetraploid tilapia as a route to the production of 100% triploids following crossing of the tetraploid with wild type. Evidence for tetraploidy induction has been obtained using both erythrocyte nuclear diameter and tetraploid chromosome spreads (see Fig. 4). However, production of tetraploid tilapia has proved difficult and seems to us on present evidence to have little commercial future.

3.4.2. Transgenic reversible sterility

To an increasing extent it is becoming clear that, unless total physical containment is available, the benefits of transgenic manipulation in fish will only be realised if an entirely ‘watertight’ sterility can be induced and used together with the transgenic improvement, Maclean and Laight (2000). Such sterility would of course have to be reversible, otherwise the sterility would involve an equivalent to ‘killing the goose that lays the golden eggs’. Can transgenesis itself be used to produce reversibly sterile tilapia fish? We believe that the answer is affirmative.

Two different levels of approach are possible with theoretical reversible sterility induction. One is to ‘knock-out’ the genomic copies of a gene whose product is crucial to sexual development and gonad formation, reversibility being accomplished by exposure of potential brood stock to the gene product so that fertility can be restored.

The alternative is to ‘knock down’ the expression of such a gene by antagonizing the specific message via transgenic induction of antisense or ribozyme against the target message.

Are there potential targets for such transgenic approaches? It seems that there are at least two, namely gonadotropin releasing hormone (GnRH) or luteinizing hormone (LH). In fish the hormonal cascade which controls gonadal development and gametogenesis involves the synthesis of GnRH in the brain and delivery of this hormone to the gonadotropes which synthesise gonadotropin. Fish GnRH is a monomer of at least three types in fish. In tilapia and related fish, the evidence is that the ‘serine 8’ or ‘sea-

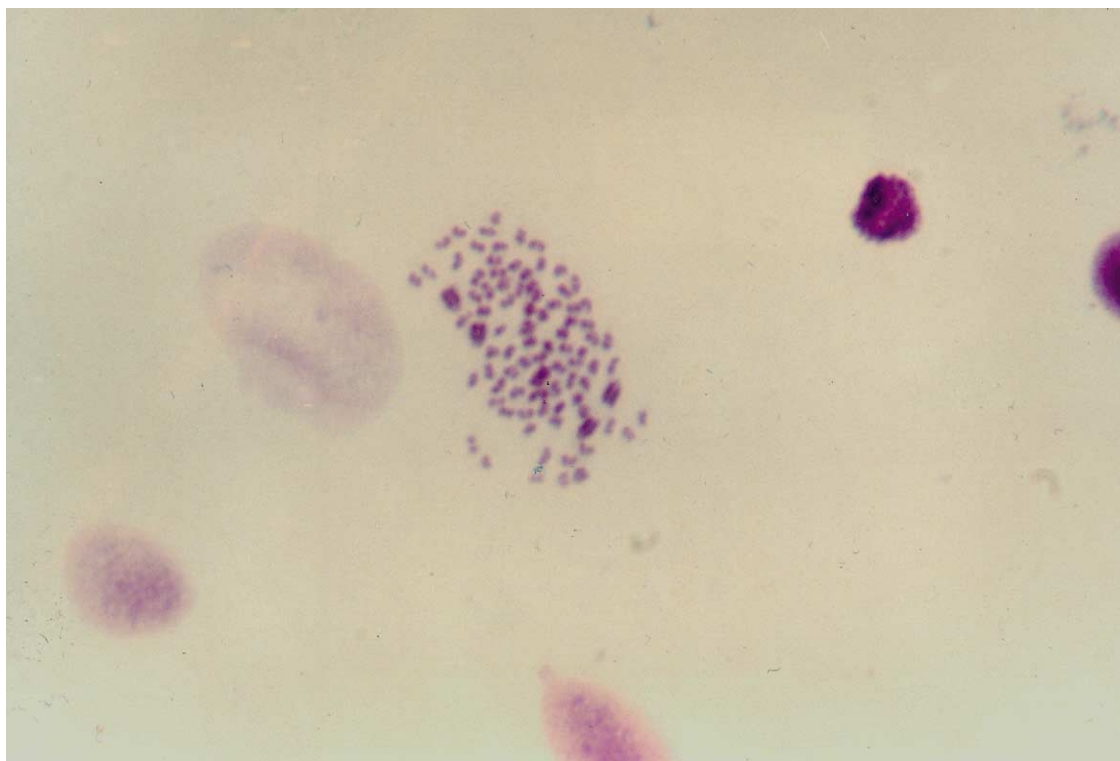


Fig. 4. Photograph of mitotic chromosome spread of presumptive tetraploid tilapia embryo following squashing ($2n = 44$) magnification $\times 400$.

bream' type GnRH, synthesised in the hypothalamus, is the form which controls gonadotropin release. However, another GnRH form, the tryptophan 7 or salmon type appears to have some direct influence also, perhaps partly as a result of its synthesis in the gonad itself (White and Fernald, 1998). Thus it is theoretically attractive to attempt to knock out or knock down production of one or another or both sorts of GnRH, and supply the hormone itself by food or injection in order to reverse the sterility.

It is of course gonadotropin itself which controls the gonad development. This hormone is a polypeptide dimer and exists in two forms, I and II. Type II is responsible for gonad development and this is the form now styled LH. The alpha and beta subunits are the products of two different genes and so either gene is a candidate target for knock out or knock down.

The conclusion from the above discussion is that both salmon type GnRH and sea bream type GnRH make possible targets in tilapia, but that the 'sea bream type' is the more likely to achieve success. In addition, tilapia LH (gonadotropin type II) is a good target in either of its subunits. We have therefore chosen to target all three tilapia genes and gene products, that is both of the above GnRH genes, and also the tilapia LH beta subunit gene. In all cases fertility could be recovered by hormonal administration to chosen potential brood stock fish. This follows the model of the hypogonadal mouse (Seeburg et al. 1987), in which a mutation in a GnRH gene conferred sterility but administration of exogenous hormone recovered fertility.

3.4.3. Embryonic stem cells and gene knock out

The success of the transgenic mouse system owes much to the availability of embryonic stem (ES) cells which allow in vitro gene targeting by homologous recombination, followed by cell selection and subsequent use of such selected cells to produce chimeric embryos and eventual non-chimeric transgenic lines. Sadly no such ES cell lines are available in tilapia, or indeed in fish of any species, although in both zebrafish, *D. rerio*, (Sun et al., 1995) and medaka, *Oryzias latipes* (Hong et al., 1996, 1998) there is experimental promise of such cell lines being produced. The lines so far produced in these fish species appear to lack complete totipotency (Hong et al., 1998, 2000) and fail to produce new germ line cells. The lack of available ES cell lines implies that other ways of accomplishing gene knock out must be sought.

In the absence of an effective ES cell system, the only remaining avenue for gene knock out is to achieve gene targeting via in vivo homologous recombination, replacing the endogenous wild type gene with a mutated silent version. However, homologous recombination is a rare event in transgenic animal systems and requires the use of substantial lengths of isogenic sequence identical to the genomic copy to be used to flank the mutated sequence to be introduced to replace the genomic target sequence. Although it is certain that any success achieved would only involve substitution of one of the two homologous copies, subsequent breeding of homozygous transgenics would achieve the desired total gene silencing.

Only one system promises to result in efficient homolo-

gous recombination (HR) in fish, and that is to use the Rad 51 protein (Sigurdsson et al., 2001), a eukaryotic equivalent of the well known bacterial Rec A protein which has also been used effectively to promote HR in both plant and mammalian cells.

A combination of a single stranded DNA sequence designed to achieve homologous recombination with, and mutagenic knock-out of, the targeted gene, together with a protein coat of the Rad 51 protein plus replication protein A (RPA), holds promise of successful gene knock out in fish. We are therefore in process of injecting tilapia eggs with such a transgenic complex, designed initially to target the tilapia serine 8 GnRH gene.

Two probes of 500 and 1000 base pairs have been designed. The two probes carry six mutations which introduce five stop codons into the GnRH coding sequence, effectively destroying the GnRH gene sequence (the wild type sequence of the GnRH extends to ten codons in total). The last mutation aims to introduce a restriction site for screening purposes. Before injection into the tilapia eggs following the methods developed in this laboratory, the probes are made single stranded by boiling and are then mixed with the recombinase (kindly provided by Dr P. Sung, Austin, USA) following the protocol described by Sigurdsson et al. (2001). After hatching the eggs are screened by PCR to detect insertion of the modification into the GnRH locus.

3.4.4. Gene knock down via RNA interference

As stated above, the alternative to gene knock-out is gene knock-down, that is to silence the expression of the target gene by antagonising the specific messenger RNA. One possible approach is to use RNA interference or RNAi. This was first reported by Fire et al. (1998) in *Caenorhabditis elegans*, where introduction of specific double stranded RNA was found to inhibit the activity of genes containing homologous sequences. This interference is post transcriptional and appears to be highly effective in *C. elegans* and *Drosophila* (Misquitta and Paterson, 1999) and mice (Wianny and Zernicka-Goetz, 2000).

There is a report of specific gene knock down in zebrafish but also of the induction of non-specific defects in zebrafish using this approach (Zhao et al., 2001). Since we already had lines of transgenic fish possessing and expressing copies of a lacZ reporter construct, we have tested RNA interference by muscle injection of dsRNA directed against the lacZ gene. However, as detailed below, we have not been able to detect any reduction in the intensity of the lacZ expression following this treatment.

A 3 kb fragment from the bacterial lacZ gene was subjected to PCR and cloned into pGEM-T. Then each strand was in-vitro transcribed using SP6 and T7 primers and the RNA transcript annealed to provide the dsRNA.

Fish of the C118 line, carrying 34 copies of lacZ in a single concatamer, were subjected to intra muscular injections of the dsRNA at marked positions above the lateral line. Fish were injected with TE buffer alone, or sense strand, antisense

strand or dsRNA (2–4 µg RNA in each case). After 48 h, muscle blocks of approx. 10 mm³ were removed from the injected sites and tested for lacZ expression. No obvious differences in expression level were detected.

Repeat experiments were carried out in which individual fish were injected in similar sites on left or right side of the body with two out of the four compared situations, but again no differences in lacZ staining were evident.

These experiments with muscle injection were followed up by egg injection with 10⁶ copies of either the sense, antisense or dsRNA and MUG assays (Rahman et al., 2000) to measure levels of lacZ expression carried out on embryo homogenates after 24 and 48 h. As seen in Fig. 5, there was no difference in the levels of expression (only 50% of embryos are expected to be transgenic since they were produced by crossing a hemizygous lacZ expressing fish of the C58 line (with 330 copies of lacZ gene) with a wild type). The proportion of embryos expressing is somewhat variable but none of the negatively expressing fish injected with dsRNA were positive for the transgene on PCR.

3.4.5. Gene knock down via antisense

Antisense sequences have now a long history of use in transgenic organisms, especially plants. A useful review of their use in animal systems is that of Sokol and Murray (1996). In general it appears that antisense is most effective against relatively rare messenger RNA species (Branch, 1998). It is also imperative to use a strong promoter in the antisense construct, and of course this must be active in the cell type in which the target gene is transcriptionally active. In our laboratory, antisense constructs have been prepared against three target genes in tilapia, namely salmon type GnRH, sea bream type GnRH, and LH beta subunit. All of these genes are expressed in areas of the brain, and in addition salmon type GnRH is expressed in gonad (testis) and sea bream type GnRH in spleen (White and Fernald, 1998). There is some logic in driving an antisense construct with the promoter sequence which regulates the target gene, and this we are doing in some cases (see Fig. 6). An alternative strategy is to use a strong ubiquitous promoter to ensure maximal expression in the target tissue. This strategy is also being followed. The following is a list of the antisense constructs currently in use in our laboratory.

- (a) Tilapia beta actin promoter/tilapia LH beta subunit gene antisense.
- (b) Carp beta actin promoter/tilapia salmon type (trypt. 7) GnRH gene antisense.
- (c) Tilapia beta actin promoter/tilapia sea bream type (serine 8) GnRH gene antisense.
- (d) Tilapia serine 8 promoter/tilapia serine 8 GnRH gene antisense.
- (e) Tilapia histone 3 promoter/tilapia serine 8 GnRH gene antisense.

Constructs (d) and (e) are not yet tested.

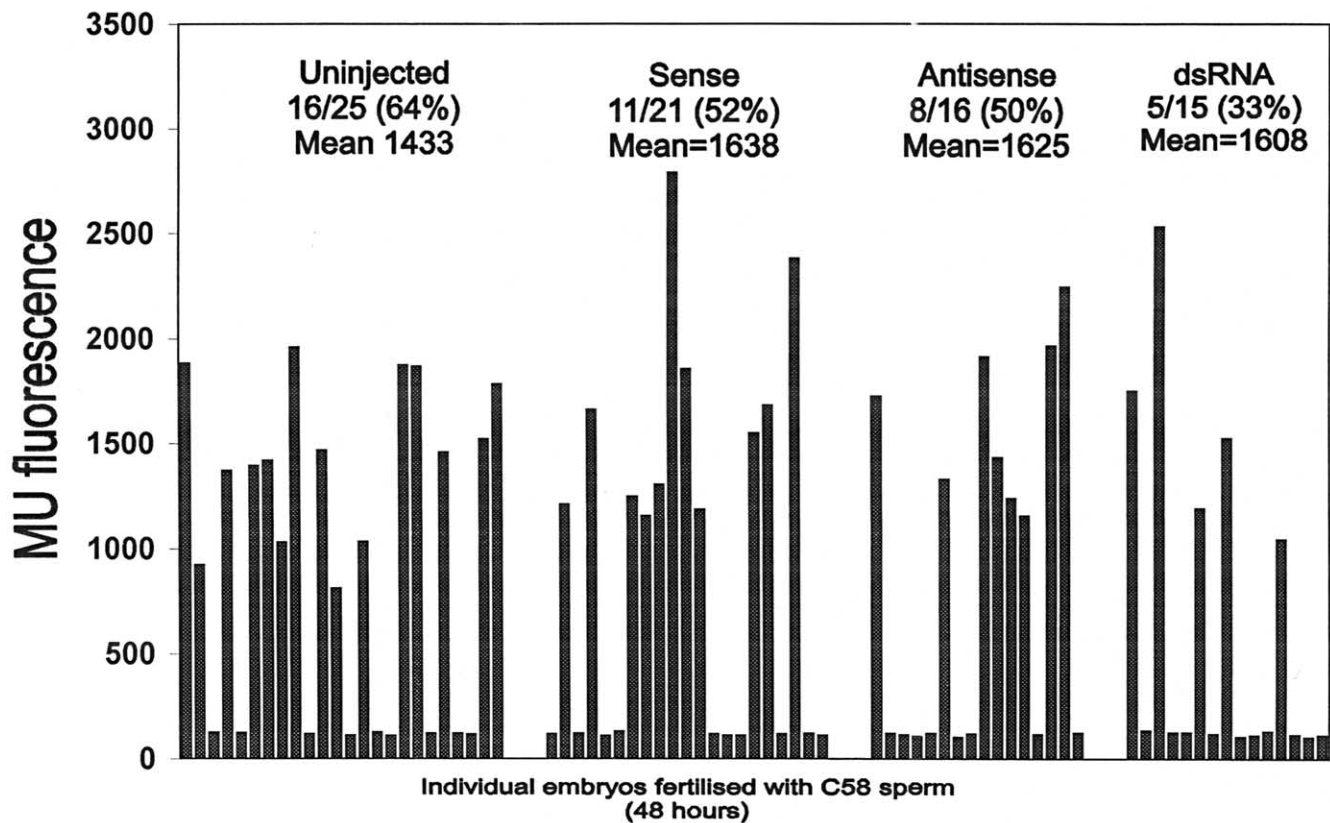


Fig. 5. Effect of RNA interference against lacZ. Histogram in which each bar represents the level of transient expression of lacZ as determined by the MUG assay in individual tilapia embryos. Since fertilised eggs resulted from eggs of wild type female fertilised with sperm of C58 male, only 50% of embryos will be transgenic and will express lacZ. Four batches of embryos are scored, the first lot grown from uninjected eggs, batch 2 injected with sense-strand RNA, batch 3 with antisense strand RNA and batch 4 with ds RNA, the RNA being directed against the lacZ message. $N = 25, 21, 16$ and 15 , respectively. There is no evidence of RNA interference with the lacZ expression.

To date only preliminary results are available. It is obvious that these constructs cannot readily be tested in tissue culture since the specific brain cell lines are not available. Expression of the tilapia serine 8 GnRH promoter has been tested with a reporter gene in spleen cells. Also transient expression in embryos is uninformative since the rele-

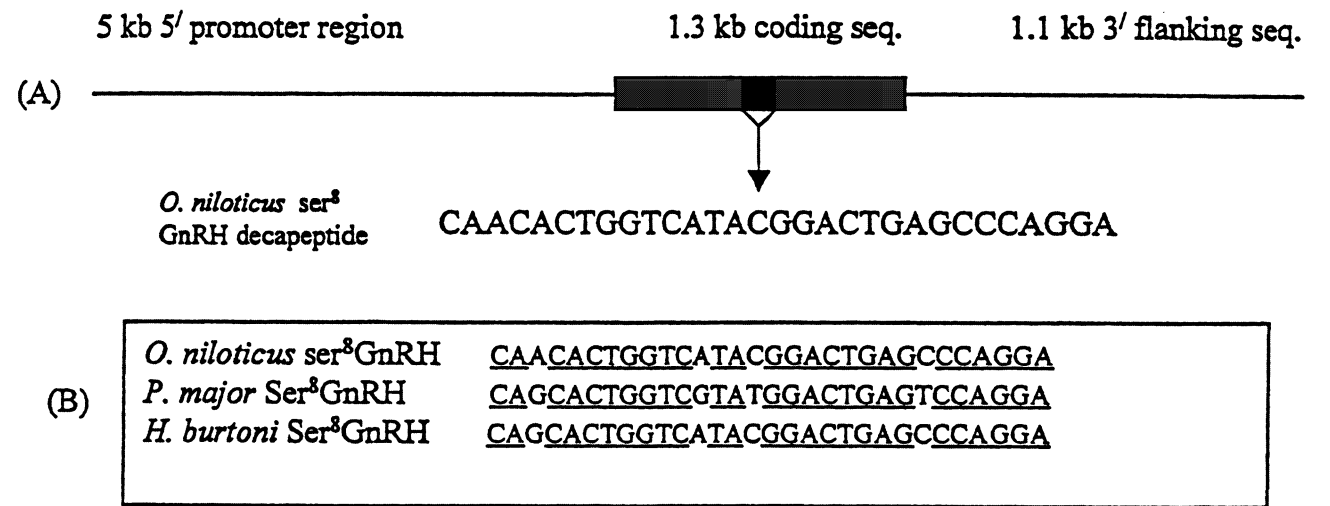


Fig. 6. Diagram giving details of the tilapia serine 8 GnRH decapeptide sequence and adjacent regions isolated in the Southampton laboratory.

vant messages are not expressed until later in development. Assay for effectivity is also tricky, since all fish will be hemizygous and the first generation will be hemizygous mosaics, and so complete effectivity may not be apparent until non-mosaic homozygous transgenics can be produced.

It is significant that the transgenics produced with construct (b) show greatly reduced fertility and in some cases total sterility, and so there is some evidence here of antisense activity. It is possible that homozygous fish for this construct will prove entirely sterile.

3.4.6. Gene knock-down via ribozyme

Ribozymes are RNA molecules with catalytic activity capable of facilitating the cutting of specific messages at target sites. They occur naturally in association with self splicing group I introns in cells such as *Tetrahymena* (Lewin, 2000). Several types of ribozymes have been described to show endoribonuclease activity. We have decided to use the hammerhead ribozyme type because it is the most commonly used in mammals. Hammerhead ribozymes cut their targets at sites which are triplets of the sequence NHH, where N can be any base and H can be A, C or T, but not G. However, all sequences are not cut with equal efficiency. The sequence that is most sensitive to ribozyme hydrolysis is GUC (guanine, uridine, cytosine triplet), and this is also the only target sequence found so far to be involved with naturally occurring hammerhead ribozyme activities. One of the main restrictions in using ribozymes is the determination of the best target site on the messenger to knock out. In the case of the serine 8 GnRH sequence this difficulty is somewhat simplified since only five GUC codons are present in the sequence of the messenger so reducing the number of potential targets to be tested. We have prepared ribozymes against all five of these sites. There is also evidence that promoters specific to type III RNA polymerase are best, and the one being tested in our laboratory is a U6 promoter.

3.5. Future exploitation of transgenic tilapia

A number of attractive avenues for future work are being explored. One of these is the production of growth enhanced lines of transgenic tilapia which are transgenic only with respect to gene sequences of tilapia origin, i.e. autotransgenics. The construct in use is shown in Fig. 3. The coinjection strategy used in our existing growth enhanced fish (Rahman et al., 1997) will also be avoided.

3.5.1. Use of transgenic tilapia as biofactories for production of human pharmaceuticals

The exploitation of transgenic organisms for the production of valuable vaccines and other pharmaceutical products is now far advanced, and examples include work with sheep, pigs, goats and poultry. For some years we have been exploring the benefits of transgenic tilapia to this end. Using a construct consisting of a human clotting factor

VII cDNA driven by a CMV promoter, we have been able to demonstrate transient expression of the protein in transgenic zebrafish and catfish, and to show that the expressed protein does indeed clot human cells 'in vitro' (Muller et al., unpublished). Through a collaboration with Aquagene Inc. of Alachua, FL, USA we are now in process of producing lines of transgenic tilapia expressing human factor VII driven by a tilapia tissue specific promoter and purifying the protein from the tilapia tissues. Germ line transmission of the construct has already been achieved and we are now testing the quality and quantity of protein produced.

The particular attractions of the tilapia model in this regard are: (a) comparatively low cost; (b) short generation time and therefore a fairly short timeframe between choice of product and first production; (c) comparative ease of making and producing transgenic tilapia; and (d) public perception of fish as 'lower' vertebrates and thus more acceptable as biofactory systems than birds or mammals.

3.6. Future exploitation of the tilapia genome

Both on grounds of consumer acceptance of products and effectiveness of regulatory sequences in terms of comparative homology, it is clearly advantageous to isolate and use sequences of tilapia origin whenever possible. Although the mining of the tilapia genome is at a very early stage, the following promoter sequences of tilapia origin are already in use in our laboratory.

Promoters:

H3 histone;
serine 8 GnRH;
L18 ribosomal protein (obtained from Molina et al. Liege);
HSP 70;
beta actin;
vitellogenin (obtained from Aquagene);
U6 small nuclear RNA.

With all of these promoters it is desirable to optimise the length of upstream sequence in order to maximise expression in a construct. This optimisation can be difficult if a relevant cell line is not available. With the serine 8 type GnRH promoter, there is the fortunate observation that this species of GnRH is expressed in spleen as well as in brain hypothalamus (White and Fernald, 1998). We have therefore used primary cell cultures of tilapia spleen to test different lengths of this promoter when in combination with a lacZ reporter. The expression obtained is specific to spleen cells. Fig. 6 shows some detail of the sequence so far isolated.

Tilapia coding sequences so far isolated include:

growth hormone (GH) (Liege laboratory);
serine 8 GnRH;
tryptophan 7 GnRH;
LH.

Acknowledgements

We are grateful for financial support to the DFID Fish Genetics Programme for the benefit of developing countries, to the EU for collaborative support, to the Government of Iran for a studentship to Hamid Farahmand and to Aquagene Inc. of Alachua, FL.

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